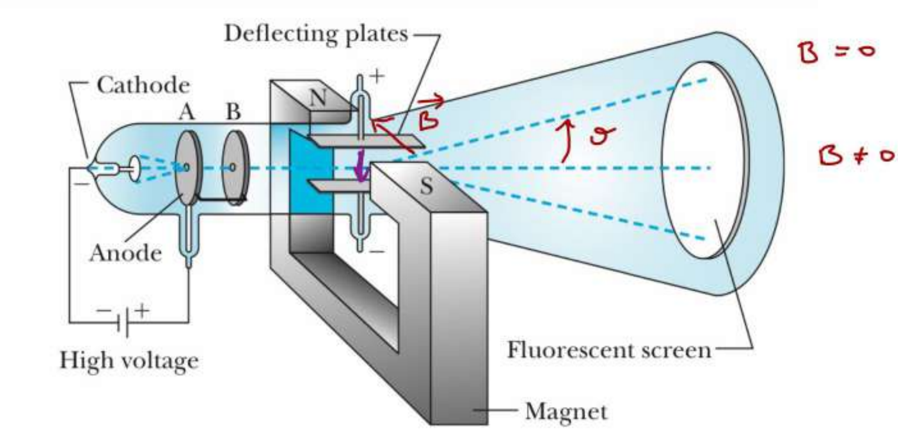
Experimental basis of quantum mechanics

The experimental basis of quantum physics

Electrons and charge quantization

Discovery of electron (Thomson)



Goal : extract q/m for particles emitted

Lorentz : $\underline{F} = q\,\underline{E} + q\,\underline{v} \times \underline{B}$

$v_x = v_0$

$B=0$: $\Delta v_y \approx \Delta t \cdot a_y = \Delta t \cdot \frac{q \cdot E}{m} = \frac{\ell}{v_0} \cdot \frac{q \cdot E}{m}$

$\Rightarrow \vartheta \simeq \frac{v_y}{v_0} = \frac{\ell \cdot q \cdot E}{v_0^2 \cdot m}$

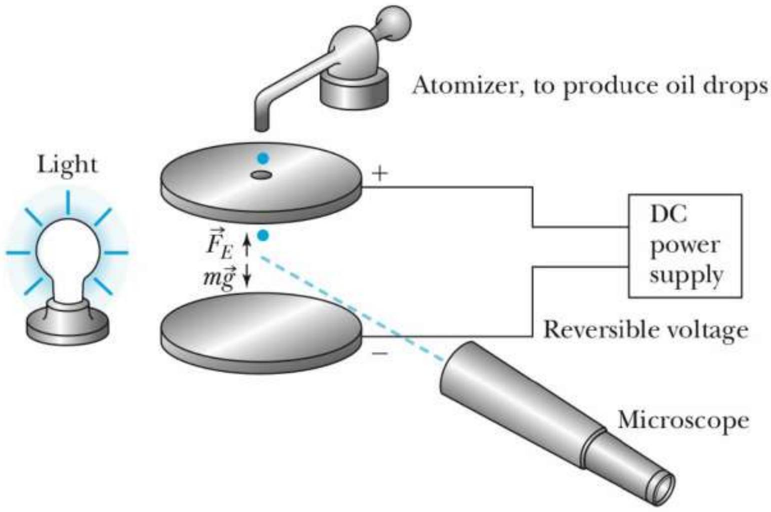
$B = \frac{E}{v_0} \Rightarrow \vartheta = 0 \Rightarrow v_0 = \frac{E}{B}$

$\rightarrow \frac{|q|}{m} = \vartheta\,\frac{v_0^2}{\ell \cdot E} = \vartheta\,\frac{E}{B^2 \cdot \ell}$

large!

negatively charged

Millikan experiment



1. stop oil drops (negatively charged)

$$mg = q \cdot E = q \cdot \frac{V}{d} \qquad (1)$$

2. $$m = \frac{4\pi}{3} R^3 \cdot \rho \qquad (2)$$

3. drop it to determine R ...

$$m \cdot g = 6\pi \cdot \eta \cdot R \cdot v \qquad (3)$$

measure this

viscosity

$\Rightarrow$ determine R: $\frac{4\pi}{3} R^3 \rho \cdot g = 6\pi \, \eta \, R \, v$

(3) $\Rightarrow$ $R = \frac{3}{\sqrt{2}} \left(\frac{\eta \cdot v}{\rho \cdot g} \right)^{1/2}$

(2) $\Rightarrow$ m

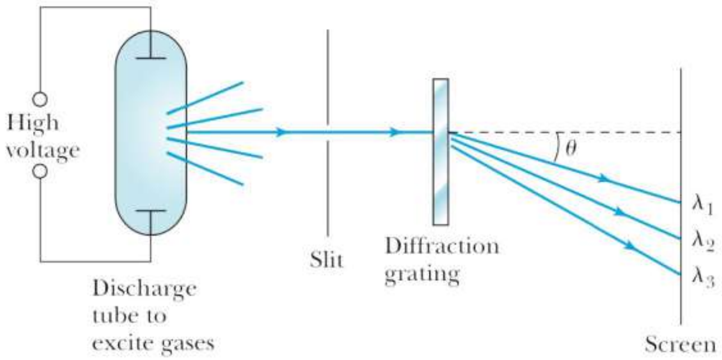
(1) $\Rightarrow$ q $\Rightarrow$ discrete $q = -e$

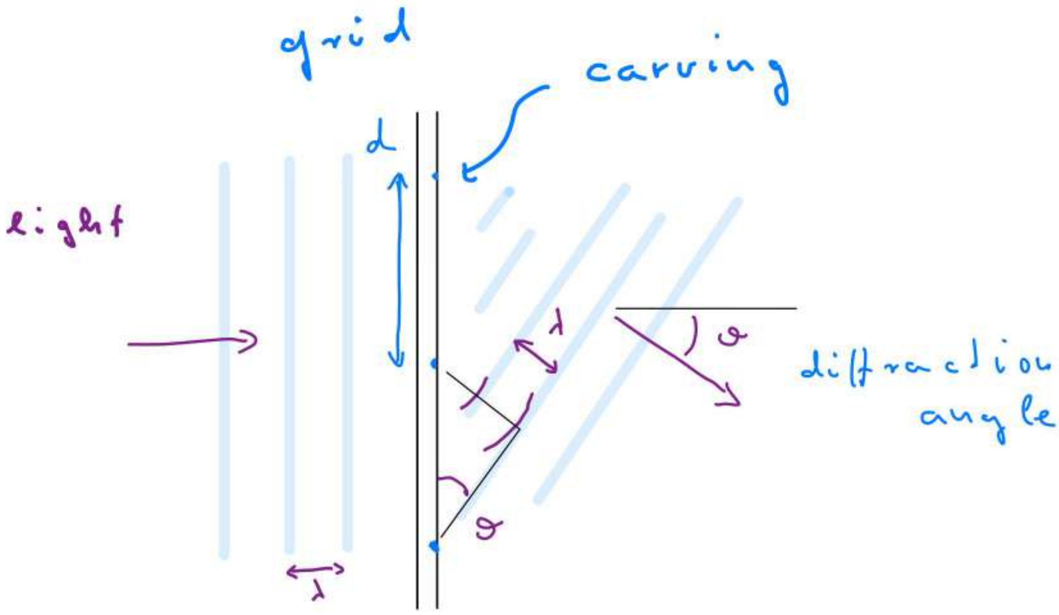
$e \cong 1{,}6 \cdot 10^{-19}$ C

Electromagnetic radiation, quanta, and photons

Line spectra

diffraction → light is wave ...





$$d \cdot \sin(\vartheta) = n \cdot \lambda$$

order of diffraction
strongest: $n = 1$

line spectrum:

atomic spectrum consists of lines

H has particularily simple spectrum:

- four visible lines (Balmer series)

$$\frac{1}{\lambda} = R_H \left(\frac{1}{4} - \frac{1}{k^2} \right) \qquad k = 3, 4, 5, 6$$

- infrared lines (Paschen)
- ultraviolet lines (Lyman)

$\Rightarrow$ Ritz, Rydberg

$$\frac{1}{\lambda} = R_H \left(\frac{1}{n^2} - \frac{1}{k^2} \right)$$

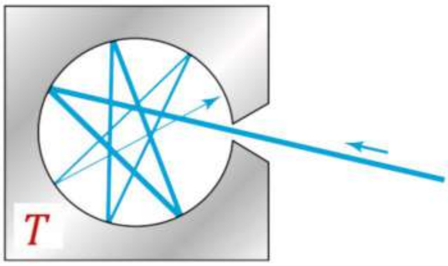
Rydberg constant $R_H = 1.096 \cdot 10^7 \frac{1}{m}$

$= 10,96 \, (\mu m)^{-1}$

Lyman: $n = 1$

Black Body radiation

oven



black body
(absorbs light)

- color of hot metal $\Leftrightarrow$ T only
- investigate outcoming spectrum
- power of radiation coming out

- color $\lambda_{max} \cdot T = 2.89 \cdot 10^{-3}\ m \cdot K$

  Wien

- power $P \propto$ radiating surface $\cdot\ T^4$

$$R \equiv \frac{P}{A} = \epsilon\ \sigma \cdot T^4$$

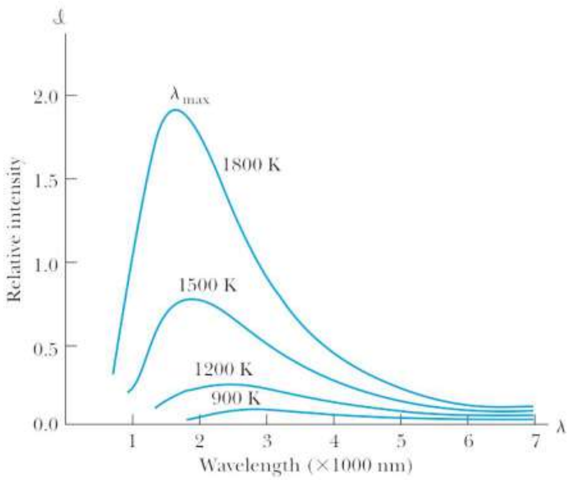
$\epsilon = 1$ for black body

Stefan - Boltzmann constant

$$\sigma = 5.67 \cdot 10^{-8}\ \frac{W}{K^4 \cdot m^2}$$

- spectrum

$S(\lambda, T)\, d\lambda \cdot dA$ = power emitted in $[\lambda, \lambda + d\lambda]$



Planck:

$$\boxed{S(\lambda, T) = \frac{2\pi c^2 h}{\lambda^5} \frac{1}{e^{hc/\lambda k_B T} - 1}}$$

Planck's constant:

$h = 6.62 \times 10^{-34}\ J \cdot s$

Boltzmann constant

$k_B = 1.38 \times 10^{-23}\ J/K$

Planck's assumption:

- Oven wall has oscillators with quantized energy $E_n = h \cdot \nu \cdot n$, $n = 0, 1, \dots$

- thermal equilibrium (Boltzmann law)

$$P_n \sim e^{-\frac{E_n}{k_B T}} = e^{-h \cdot \nu \cdot n / k_B T}$$

- electromagnetic modes in cavity are in equilibrium with oscillators, which absorb and emit energy in discrete quantum, $h \cdot \nu$

All other laws follow from it ...

- e.g. $\lambda \to \infty$

$$S = \frac{2\pi c^2 h}{\lambda^5} \frac{1}{e^{ch/\lambda k_B T} - 1} \approx \frac{2\pi c}{\lambda^4} \cdot k_B T$$

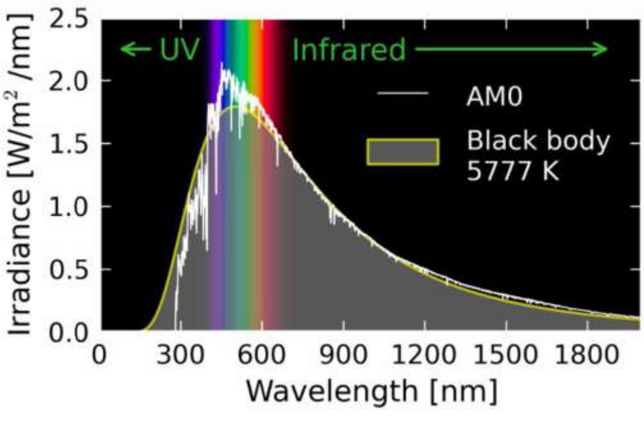
Rayleigh - Jeans

- $$\frac{\partial}{\partial \lambda} S(\lambda, T) = 0 \quad \Rightarrow \quad \lambda_{max} \cdot T = cst$$

- $$R = 2\pi c^2 \cdot h \cdot \int_0^\infty d\lambda \frac{1}{\lambda^5} \frac{1}{e^{\frac{ch}{\lambda k_B T}} - 1} = \dots = cst \cdot T^4$$

Sun

~ black body (outside atmosphere)



Derivation of Planck's law (sketch)

1) Number of modes $\omega = 2\pi \cdot \nu = c \cdot k = c \cdot \frac{2\pi}{\lambda}$

$$2 \frac{V}{(2\pi)^3} \cdot d^3k = V \frac{1}{\pi^2} \cdot k^2 \cdot dk = V \cdot 8\pi \frac{d\lambda}{\lambda^4}$$

2 modes $\quad k = \frac{2\pi}{\lambda}$

2) E.M. mode ~ oscillator of frequency ν

$$\Rightarrow \text{energy} \Rightarrow \langle E \rangle = \sum_n n \cdot h\nu \cdot P_n = \ldots = \frac{h \cdot \nu}{e^{\frac{h \cdot \nu}{k_B T}} - 1}$$

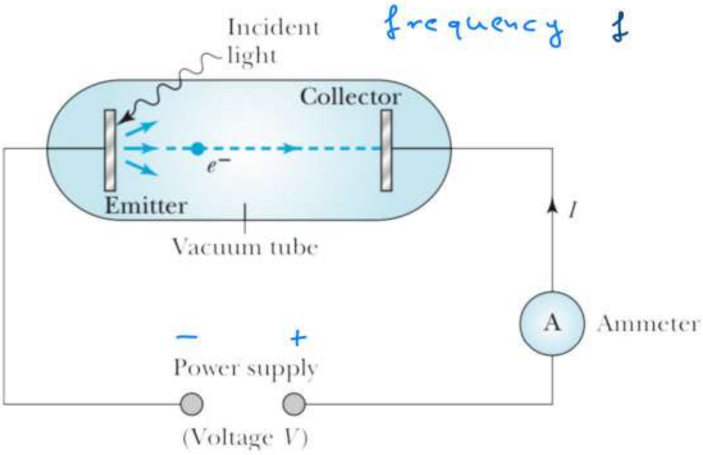
3) energy density of energy

$$u(\lambda, T)\, d\lambda = 8\pi \frac{1}{\lambda^4} d\lambda \cdot \frac{\frac{hc}{\lambda}}{e^{\frac{hc}{\lambda k_B T}} - 1}$$

4) energy stream over surface :

$$S(\lambda, T) = \frac{c}{4} \cdot u(\lambda, T) = 2\pi \frac{c^2}{\lambda^5} \frac{1}{e^{\frac{hc}{\lambda k_B T}} - 1} \checkmark$$

Photoelectric effect



frequency f

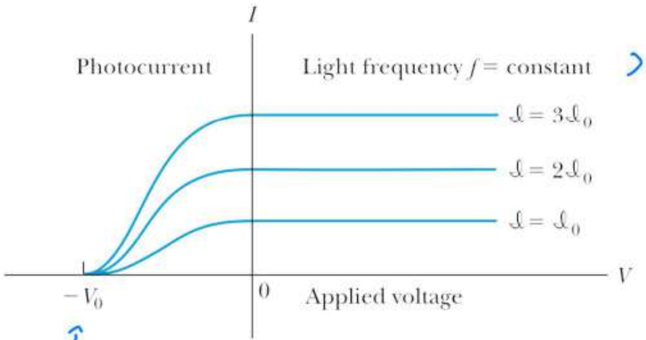
light → electric current

Einstein :

light is quantized

1. if $f < f_{min}$ ⇒ no current, even if intensity is large

2. if $f > f_{min}$ ⇒ current



$> f_{min}$

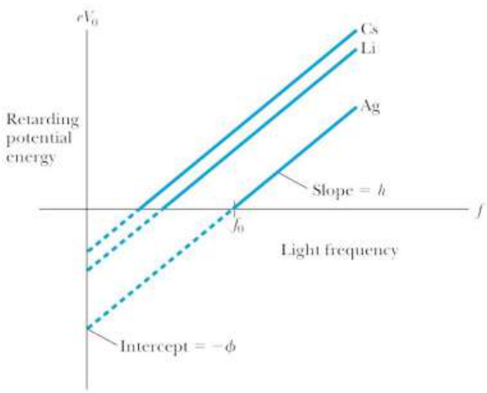
maximal kinetic energy of emitted e^- $= eV_0$

independent of intensity !

current saturates for $V>0$ (collect all e^-) ✓

proportional to intensity of light

3. $K_{max} = eV_0$ depends on f but not on $\mathcal{I}$



$$\boxed{K_{max} = eV_0 = h \cdot f - \phi}$$

material dependent (work function)

Explanation: Einstein (1905)

- energy of ligh of frequency f is quantized in units of $h \cdot f$

- removing e^- from metal costs ϕ

  if $hf < \phi$ $\Rightarrow$ energy of light is insufficient

  if $hf > \phi$ $\Rightarrow$

  electrons are kicked out with

  energy $K = hf - \phi = eV_0$

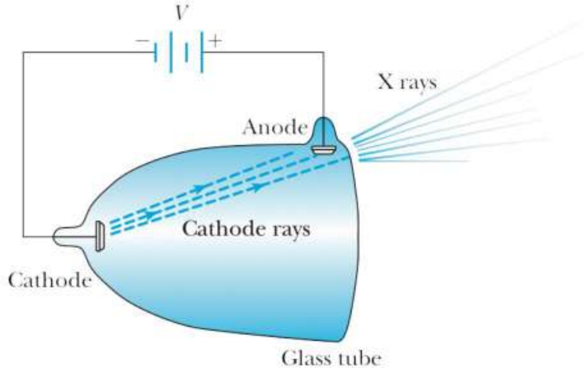
light ~ particle! $\Rightarrow$ photon

X rays

Discovery of X rays (Röntgen and Becquerel)

Röntgen:

cathode ray tube

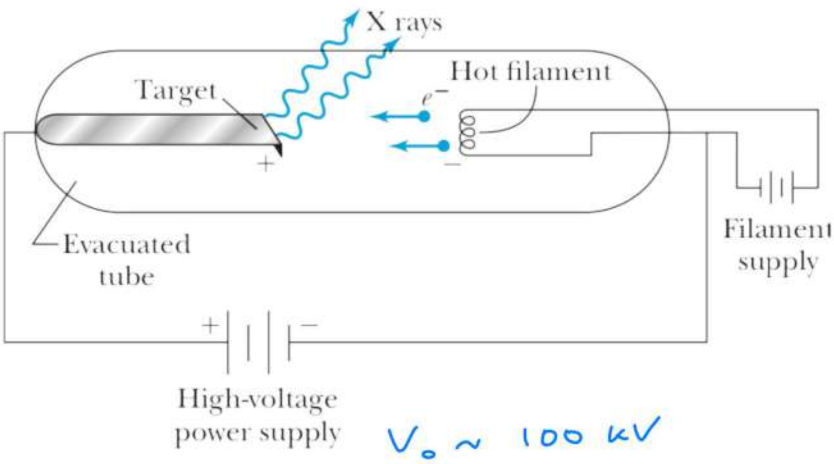


wrapped in paper

discovered picture 1m away

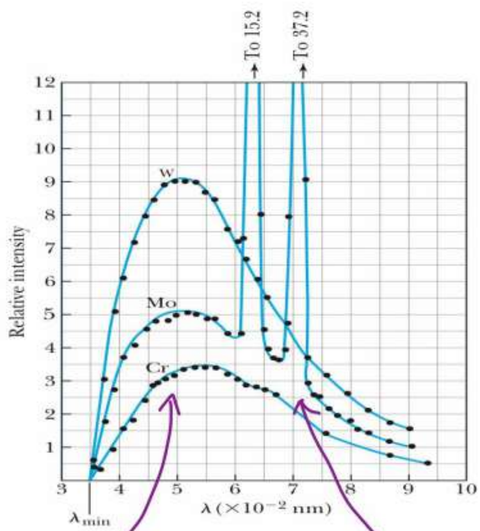
Becquerel $K_2(UO_2)(SO_4)_2$ (photoluminescent) → emitted X-ray!

X-ray sources



$V_0 \sim 100\ kV$

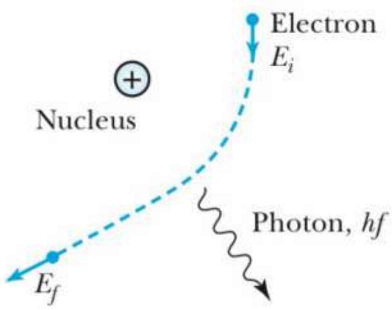
spectrum:



Bremsstrahlung

discrete lines

Bremsstrahlung



$$E_i = eV_0 = E_f + hf$$

$$eV_0 = hf_{max} = \frac{hc}{\lambda_{min}}$$

X-ray diffraction

Compton : X-rays behave as particles (photons)

Q: do they behave as waves, too ?

A: Laue , Bragg + Bragg : YES

$E = h \cdot f = 500 \text{ keV} \quad \Rightarrow$

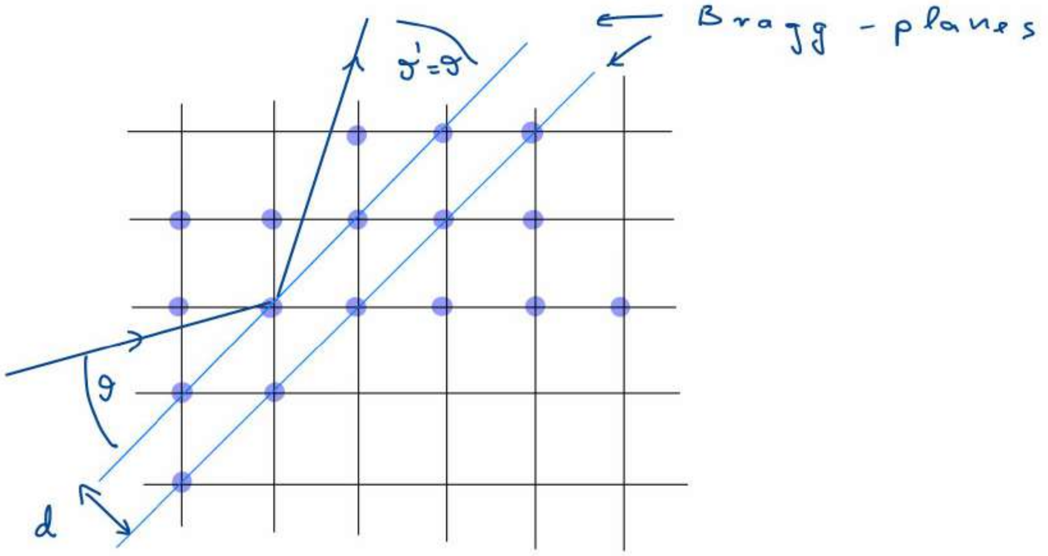
$$\lambda = \frac{c}{f} = \frac{c \cdot h}{30 \text{ keV}} = \frac{1{,}98 \cdot 10^{-25}}{3 \cdot 10^{4} \cdot 1{,}6 \cdot 10^{-19}} \text{ m}$$

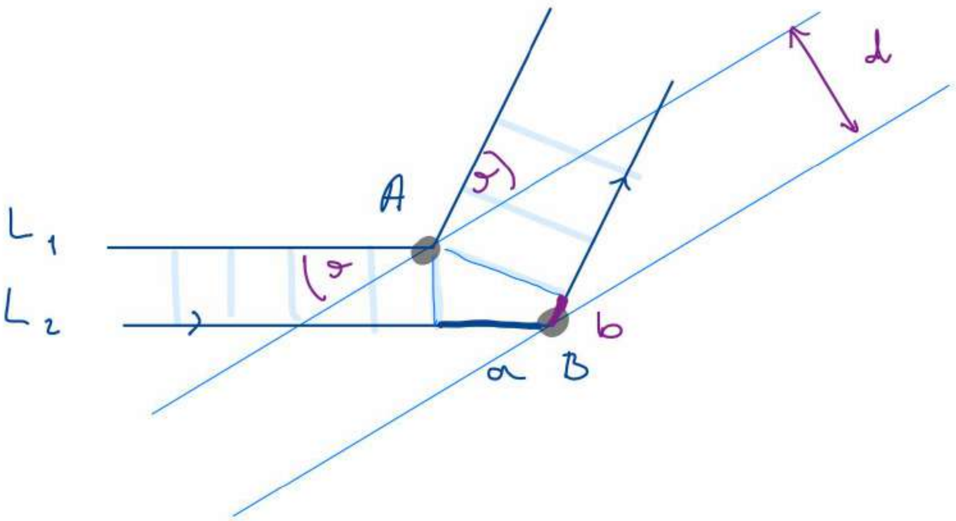
dental

$$= 0{,}41 \cdot 10^{-10} \text{ m}$$

~ atomic distance !

! use crystal lattice for diffraction !





Interference if length difference between

P1 and P2 $\qquad L_2 - L_1 \overset{!}{=} n \cdot \lambda$

claim:

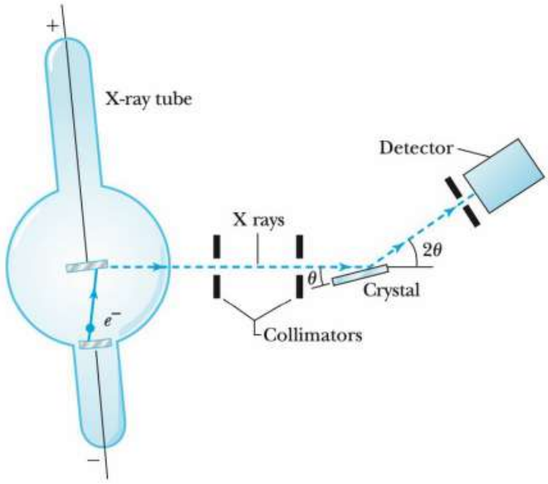
$$L_2 - L_1 = a + b \overset{!}{=} 2d \cdot \sin\vartheta$$

for $\underline{\underline{\text{any}}}$ two atoms in planes

Bragg condition

$$\boxed{2d \sin\vartheta = n \cdot \lambda}$$

Apparatus:



$\Rightarrow$ analysis of complex molecules !

(DNA !)

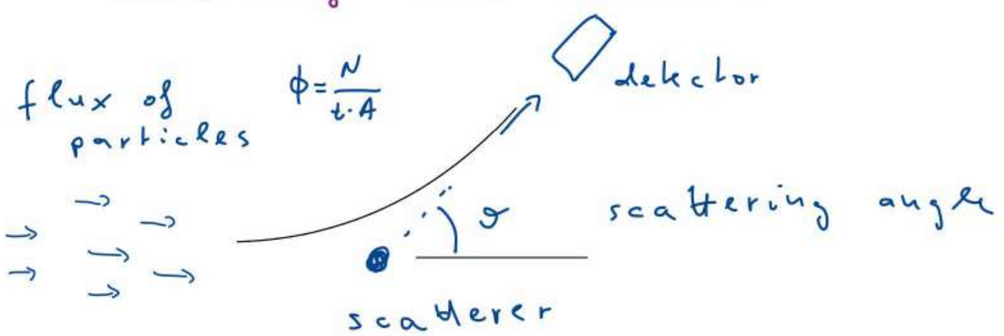
Atoms

Rutherford scattering

Coulomb scattering of α-particles on nuclei

Scattering cross section:

$\phi = \frac{N}{t \cdot A}$



particles detected $\propto \phi \cdot t \cdot d\Omega$

spherical angle = $d\Omega = d\vartheta \sin\vartheta \, d\varphi$

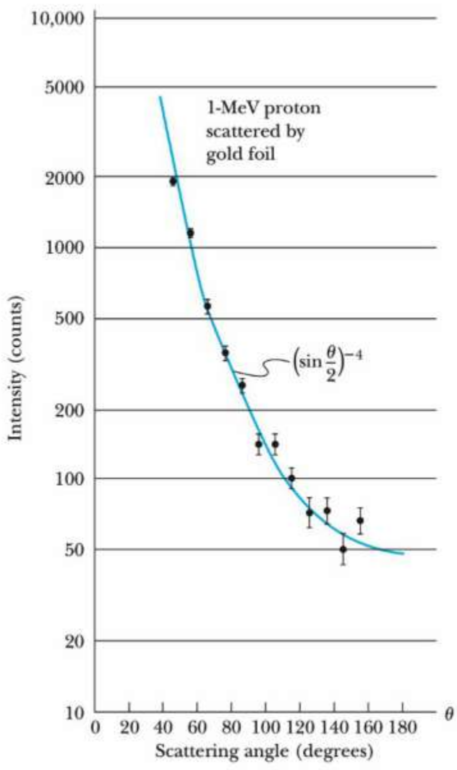
$$N_{detected} = \phi \cdot t \cdot N_{scatterer} \cdot \sigma(\Omega) \cdot d\Omega$$

differential cross section

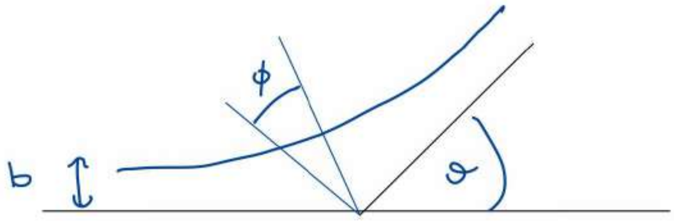
Rutherford cross-section:

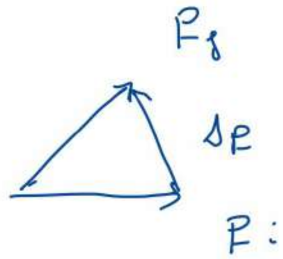
$$V(r) = \frac{z e^2}{2\pi\epsilon_0} \frac{1}{r} \quad \text{potential}$$

$$\sigma(\vartheta) : \frac{1}{4} z^2 \frac{e^4}{(4\pi\epsilon_0)^2} \cdot \frac{1}{E^2} \frac{1}{\sin^4(\frac{\vartheta}{2})}$$



Derivation (supplementary)





(1) $\Delta p = 2\,p \cdot \sin\frac{\vartheta}{2}$ $\qquad\qquad \alpha = \frac{2Ze^2}{4\pi\epsilon_0}$

(2) $F_{\Delta p} = F \cdot \cos\phi = \frac{\alpha}{r^2}\cos\phi$

$$\Delta p = \int dt\; F_{\Delta p} = \int dt\; \frac{\alpha}{r^2}\cos\phi$$

(3) angular momentum: $\underset{\nearrow \text{ impact parameter}}{b} \cdot p = m r^2 \dot{\phi}$

$$\Rightarrow \quad 2p\sin\left(\frac{\vartheta}{2}\right) = \int dt\; \frac{\alpha \cdot m \dot{\phi}}{b \cdot p}\cos\phi =$$

$$= \frac{\alpha \cdot m}{b \cdot p}\; 2\int_0^{(\pi-\vartheta)/2} \cos\phi\; d\phi = \frac{2\alpha \cdot m}{b \cdot p} \cdot \cos\left(\frac{\vartheta}{2}\right)$$

$$\Rightarrow \quad b = \frac{1}{2}\;\frac{\alpha}{E}\;\operatorname{ctg}\left(\frac{\vartheta}{2}\right)$$

$$2\pi \cdot b\, db = 2\pi \cdot \sigma(\vartheta)\, d\vartheta \quad \Rightarrow \quad \sigma(\vartheta) = b\left|\frac{db}{d\vartheta}\right| =$$

$$= \frac{1}{16}\;\frac{\alpha^2}{E^2}\;\frac{1}{\sin^4(\vartheta/2)}$$